

नमस्कार मुंबई!



MCP
Dev Summit
Mumbai

Presented by:



Agentic AI
Foundation

14-15 June 2026

MUMBAI, INDIA
#MCPDEVSUMMIT



Beyond Tool Calls

Unlocking Interactive, Token-Smart Agents With MCP Apps



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\$WhoAml

I am Suraj

A senior engineer at HDFC Bank

- Responsible for the Multi-Factor Authentication (MFA) system
- Leading retail banking MCP Server developments



Connect on LinkedIn

Outside work, I am contributing to the Go implementation of the MCP spec

0 Open

✓ 5 Closed

Author

Label

Projects

Use omitzero json tag for byte array field in ResourceContents, instead of omitempty

✓

#782

by IAmSurajBobade

Contributor

was merged on Feb 9

• Approved

chore: use rand.Text and slog.DiscardHandler over intrernal implementation

✓

#773

by IAmSurajBobade

Contributor

was merged on Feb 3

• Approved

Implement SEP-973

✓

#570

by IAmSurajBobade

Contributor

was merged on Nov 13, 2025

• Approved

SSE: Add support for SSE handler options

✓

#508

by IAmSurajBobade

Contributor

was merged on Sep 19, 2025

• Approved

mcp/server: enable server logging

✓

#501

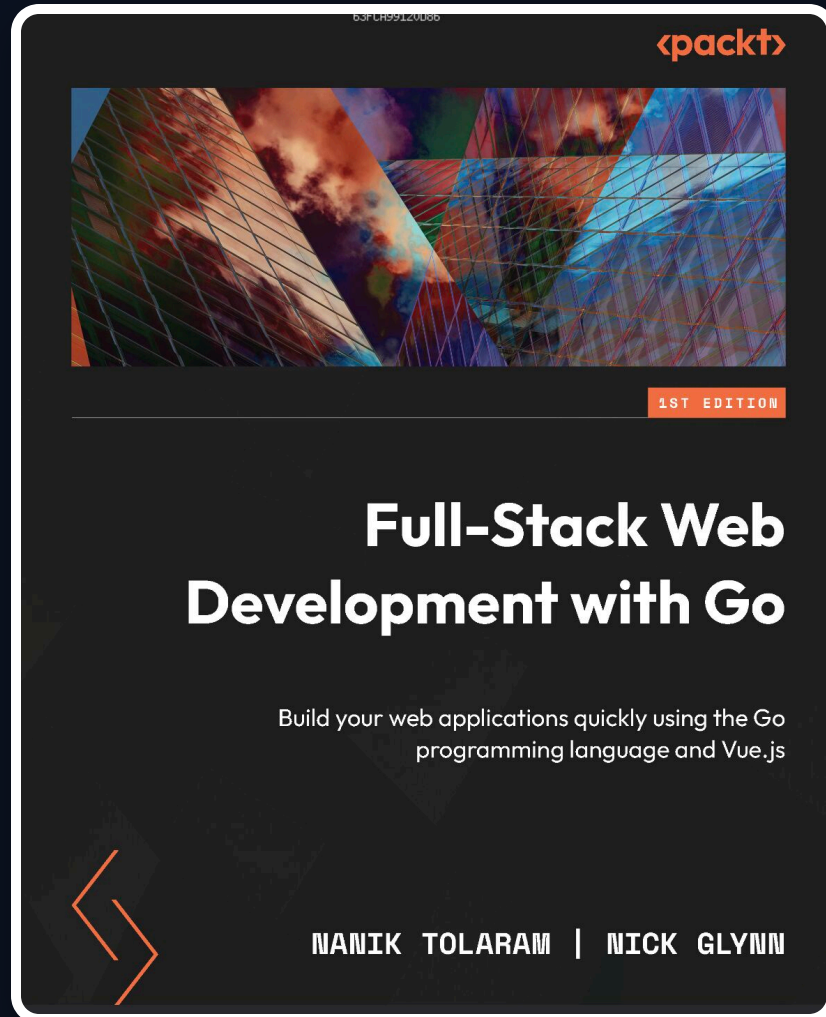
by IAmSurajBobade

Contributor

was merged on Oct 3, 2025

• Approved

Also, worked as a technical reviewer for Packt



About the reviewers

Pablo David Garaguso was born in Mar del Plata, Argentina. He graduated with two degrees in computer sciences and enterprise systems from CAECE University and later on received an MBA from CEMA University in Buenos Aires, Argentina. He has worked as an attached professor, project leader, and international manager, and currently occupies a position as a solutions architect in R&D for FamilySearch Int. Europe. Based in Finland, he has published a variety of books according to his many interests, from novels and fiction to software engineering. His latest book, *Vue 3 Applications and Design Patterns*, will be published by Packt in 2023.

Suraj Bobade is an experienced software professional, currently located in Pune, India. He completed a B.Tech in computer science from Walchand College of Engineering, Sangli.

He is passionate about software development with a keen interest in product management. He builds user-first feature-rich products while driving critical software and product design decisions.

Go has been his go-to choice for building the microservice backend and prototypes. Considering the simplicity and increasing adoption by the open source community, Suraj is optimistic that readers of this book will learn in-demand skills.

USEFUL, UNTIL USABLE

MCP got us far

`tools/list`, `tools/call`, resources, and OAuth made agents useful.



COMEDY OF DELEGATION

Delegate one weekend

A match, a cheap ticket, and several life choices.

Text-only delegation · 1/6

USER

Find and book a weekend match ticket in Mumbai if it is under ₹3,000



Text-only delegation · 2/6

USER

Find and book a weekend match ticket in Mumbai if it is under ₹3,000

AGENT

Got it. Checking available tickets under your budget...

Text-only delegation · 3/6

USER

Find and book a weekend match ticket in Mumbai if it is under ₹3,000

AGENT

Got it. Checking available tickets under your budget...

AGENT

Done. Booked the cheapest available ticket for ₹1,000



Text-only delegation · 4/6

USER

Find and book a weekend match ticket in Mumbai if it is under ₹3,000

AGENT

Got it. Checking available tickets under your budget...

AGENT

Done. Booked the cheapest available ticket for ₹1,000

USER

Great. Book a flight to Mumbai as well



Text-only delegation · 5/6

USER

Find and book a weekend match ticket in Mumbai if it is under ₹3,000

AGENT

Got it. Checking available tickets under your budget...

AGENT

Done. Booked the cheapest available ticket for ₹1,000

USER

Great. Book a flight to Mumbai as well

AGENT

Found a low fare. Booking the best-priced option...



Text-only delegation · 6/6

USER

Find and book a weekend match ticket in Mumbai if it is under ₹3,000

AGENT

Got it. Checking available tickets under your budget...

AGENT

Done. Booked the cheapest available ticket for ₹1,000

USER

Great. Book a flight to Mumbai as well

AGENT

Found a low fare. Booking the best-priced option...

AGENT

Done. Booked a Friday early-morning flight. All set for the match

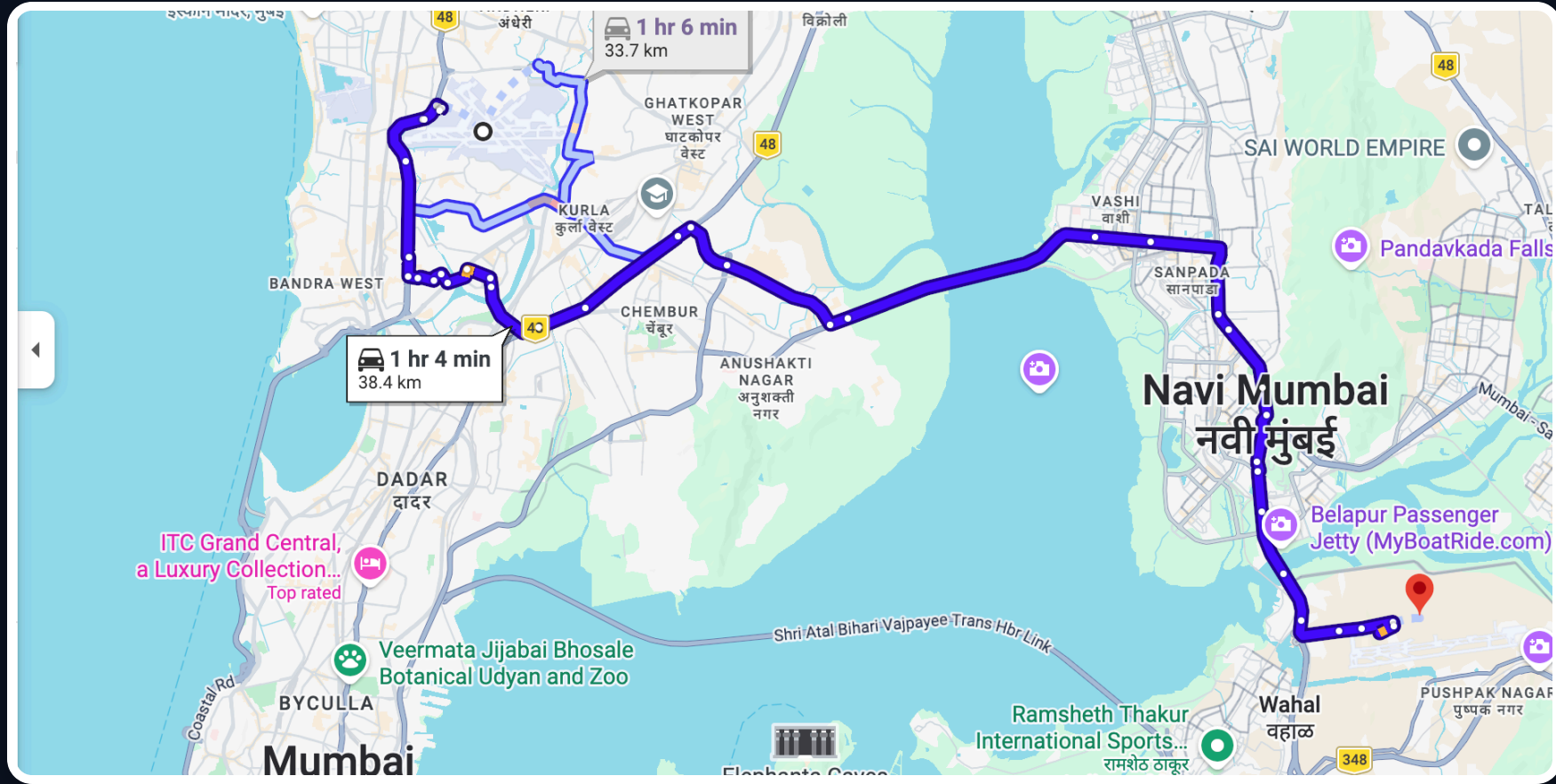


The agent did exactly what you asked

The flight is at 2 AM

You realize you never prompted for the time.

Mumbai-ish



You are sporting



Control, Uday, control



**Delegation
without visibility
is just suspense.**

FUNNY UNTIL FINANCE

Now make it banking

Same delegation pattern. Much less comedy.

Model-visible data · 1/6

USER

How much did I spend last week?

Model-visible data · 2/6

USER

How much did I spend last week?

AGENT

Calling statement tool with the default 12-month range...



Model-visible data · 3/6

USER

How much did I spend last week?

AGENT

Calling statement tool with the default 12-month range...

AGENT

You spent ₹9,000 last week. ₹6,000 was rent.



Model-visible data · 4/6

AGENT

Calling statement tool with the default 12-month range...

AGENT

You spent ₹9,000 last week. ₹6,000 was rent.

USER

How much did I spend last week?

USER

Any bills coming up?



Model-visible data · 5/6

AGENT

Calling statement tool with the default 12-month range...

AGENT

You spent ₹9,000 last week. ₹6,000 was rent.

AGENT

Fetching card transactions to infer recurring payments...

USER

How much did I spend last week?

USER

Any bills coming up?



Model-visible data · 6/6

USER

How much did I spend last week?

AGENT

Calling statement tool with the default 12-month range...

AGENT

You spent ₹9,000 last week. ₹6,000 was rent.

USER

Any bills coming up?

AGENT

Fetching card transactions to infer recurring payments...

AGENT

You have ₹13,000 SIP coming up. I also see a ₹18 transaction every day at 5 PM



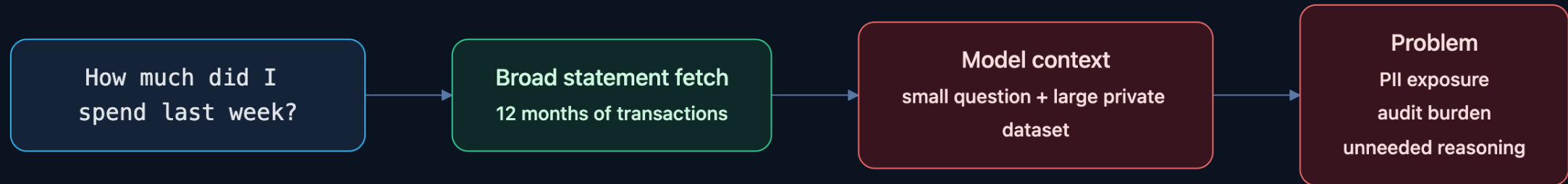
What actually broke?

No review surface:
Important choices
stayed hidden until
after execution.

Too much model
context: Small answers
pulled large private
datasets into the
reasoning path.

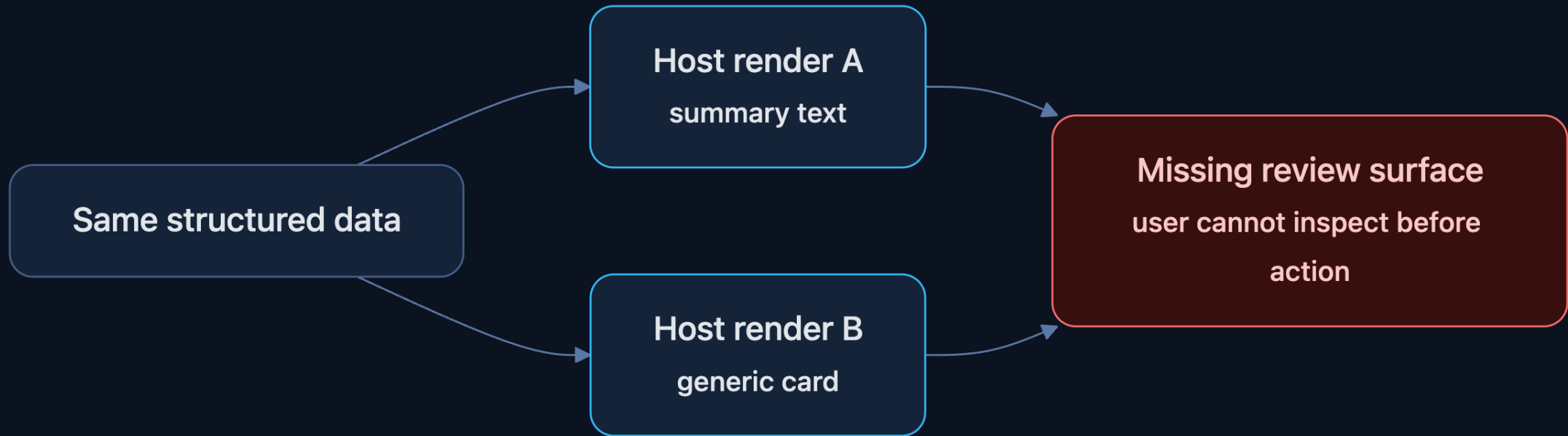
Wrong owner of truth:
Probabilistic text sat
too close to
deterministic
decisions.

Problem 1 • model sees too much



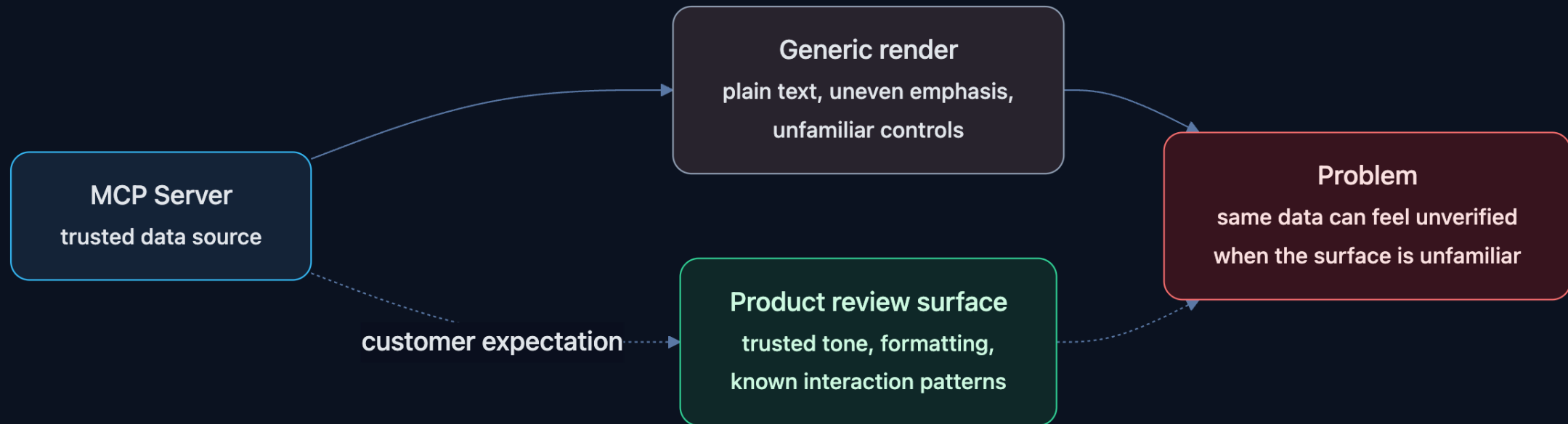
The model should see the answer context, not every private row used to compute it.

Problem 2 · user cannot inspect



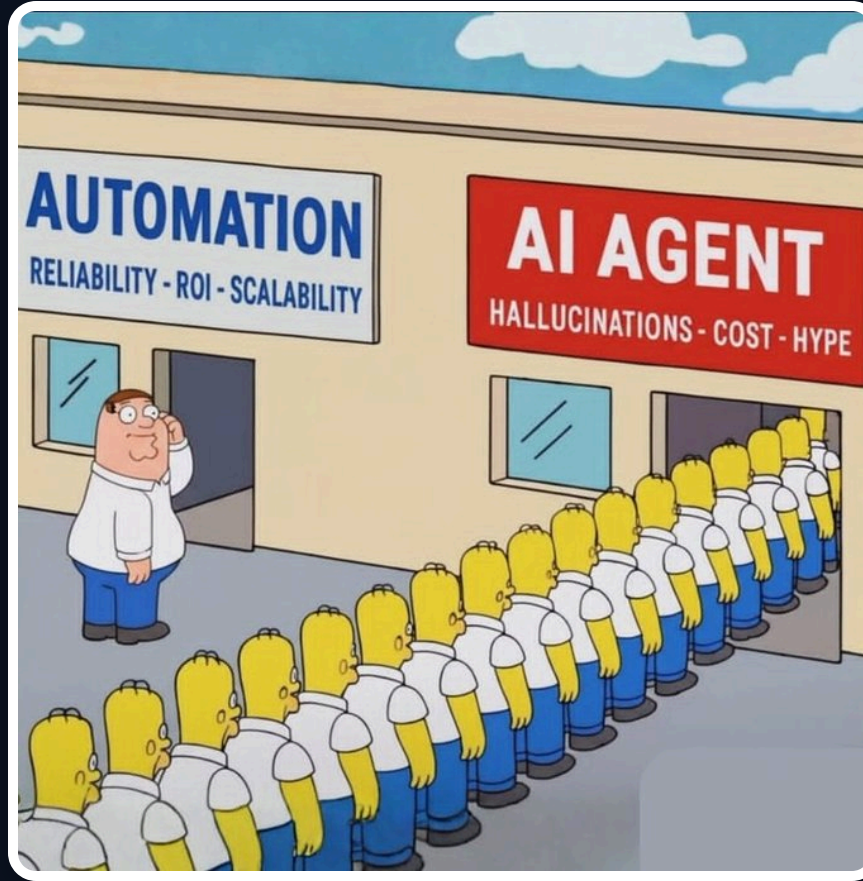
Important decisions need a review surface, not only a fluent sentence.

Problem 3 · trust surface

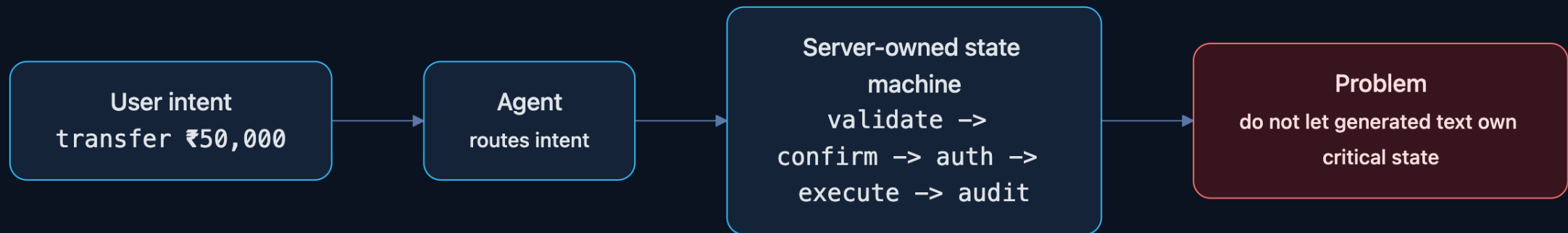


In sensitive flows, how information is rendered changes whether users trust it.

Problem 4 · irrelevant reasoning surface



Problem 5 · deterministic state



Fund transfer is a state machine, not a probability distribution.

Problem 6 • the token tax



Context-window maxing: great if you sell tokens. Less great for the customer.

**Let the UI hold
what the model
does not need to
see.**

THE UI ENTERS THE CHAT

The control surface

Interactive UI inside chat, without throwing away MCP boundaries.

What is an MCP App?

It is the first official **MCP extension** that standardizes the way to deliver **interactive UI** via MCP tools

Spec primitives

Stable spec version: `2026-01-26`

Extension ID: `io.modelcontextprotocol/ui`

Resource prefix: `ui://`

MIME type: `text/html;profile=mcp-app`

Tool-UI link: `_meta.ui.resourceUri`

Official launch: 2026-01-26. Tools declare UI metadata; Hosts render UI resources in sandboxed iframes.

What MCP Apps are NOT

- **Not** a replacement for MCP tools.
- **Not** a way to bypass auth or governance.
- **Not** a replacement for context engineering — it is a way to keep the wrong data out of context.
- **Not** required for every Host — it is progressive enhancement.

MCP Apps adds an interactive View beside the conversation, while preserving tool, resource, and auth boundaries.

TOKENS HATE LEFTOVERS

Token savings

Tool visibility controls

```
{
  "tools": [ {
    "name": "awesome-tool",
    ...
    "_meta": { "ui": {
      "resourceUri": "ui://awesome-resource",
      "visibility": ["app", "model"]
    }}
  ]
}
```

<code>_meta.ui.visibility</code>	View Access	Model Access
<code>["app"]</code>	Yes	No
<code>["model"]</code>	No	Yes
<code>["app", "model"]</code>	Yes	Yes
omitted (default)	Yes	Yes

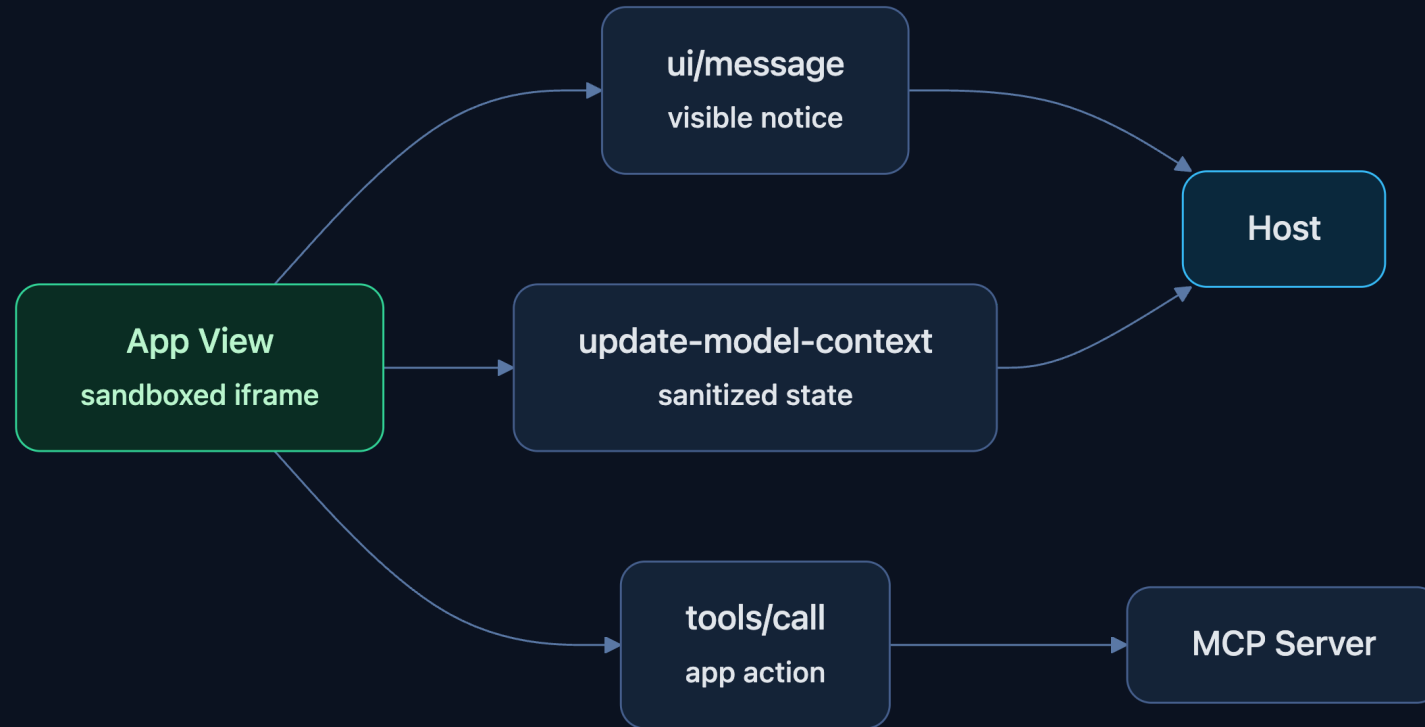
app-only tools are for buttons, filters, pagination, form submits — UI chatter the model does not need.

response fields visibility

```
{
  "content": "Statement loaded. Closing balance Rs. 2,45,000",
  "structuredContent": { "closingBalance": 245000 },
  "_meta": {
    "rows": [ "... " ],
    "renderHints": { "currency": "INR" }
  }
}
```

keep `content` minimal and verify each target host before relying on `_meta` privacy.

Bidirectional communication



Pick the rail: `ui/message`, `ui/update-model-context`, or `tools/call`.

Personalization without token theater

At `ui/initialize`, the Host shares fields like:

`theme`

`styles`

`displayMode`

`availableDisplayModes`

`containerDimensions`

`locale`

`timeZone`

`platform`

`safeAreaInsets`

Display modes

inline

MCP App View

fullscreen

Full workflow surface

pip

Floating app

Demo

MCP server · Goose host



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Beyond Tool Calls

Unlocking Interactive, Token-Smart Agents With MCP Apps

Suraj B

MCP Dev Summit Mumbai · 14-15 June 2026

▶ 0:00 / 2:36



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```
<script> (() => { const video = document.getElementById("summit-demo-video"); const
```

Total Tokens

606

Deck + references



Capability negotiation · current spec

One-time `initialize` handshake — caps live in `capabilities.extensions`.

Host → Server · initialize request

```
{
  "jsonrpc": "2.0",
  "id": 1,
  "method": "initialize",
  "params": {
    "protocolVersion": "2025-11-25",
    "capabilities": {
      "extensions": {
        "io.modelcontextprotocol/ui": {
          "mimeType": ["text/html;profile=mcp-app"]
        }
      }
    },
    "clientInfo": {
      "name": "mcp-host", "version": "1.0.0"
    }
  }
}
```

Server → Host · initialize result

```
{
  "jsonrpc": "2.0",
  "id": 1,
  "result": {
    "protocolVersion": "2025-11-25",
    "capabilities": {
      "tools": {},
      "resources": {},
      "extensions": {
        "io.modelcontextprotocol/ui": {}
      }
    },
    "serverInfo": {
      "name": "mcp-server", "version": "1.0.0"
    }
  }
}
```

Capability negotiation · RC 2026-07-28

Stateless — no handshake. Client caps ride in `_meta` on every request.

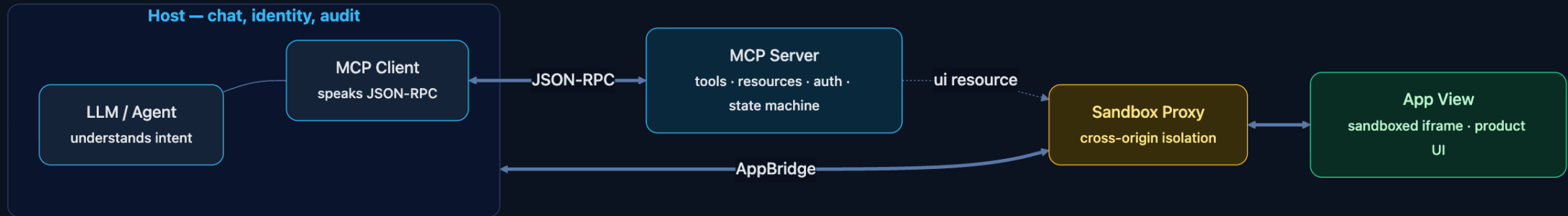
Host → Server · every request carries caps

```
{
  "jsonrpc": "2.0",
  "id": 7,
  "method": "tools/call",
  "params": {
    "name": "awesome-tool",
    "arguments": {},
    "_meta": {
      "io.modelcontextprotocol/protocolVersion": "2026-07-28",
      "io.modelcontextprotocol/clientCapabilities": {
        "extensions": {
          "io.modelcontextprotocol/ui": {
            "mimeType": ["text/html;profile=mcp-app"]
          }
        }
      }
    }
  }
}
```

Server → Host · server/discover result

```
{
  "jsonrpc": "2.0",
  "id": 8,
  "result": {
    "supportedVersions": ["2026-07-28"],
    "capabilities": {
      "extensions": {
        "io.modelcontextprotocol/ui": {}
      }
    },
    "serverInfo": {
      "name": "mcp-server", "version": "1.0.0"
    }
  }
}
```

Four moving parts



Component	Role
Host	Conversation, identity, auth tokens, audit log.
MCP Server	Tools, business logic, auth checks, <code>ui://</code> templates.
Sandbox Proxy	Cross-origin iframe boundary for web hosts.
View	Your HTML/JS in a sandboxed iframe.

The Host

1 / 4

Owns the model session and the MCP client — and renders Views

HOST claude.ai · Claude Desktop · VS Code Copilot

LLM / Agent

reasoning · tool selection



MCP Client

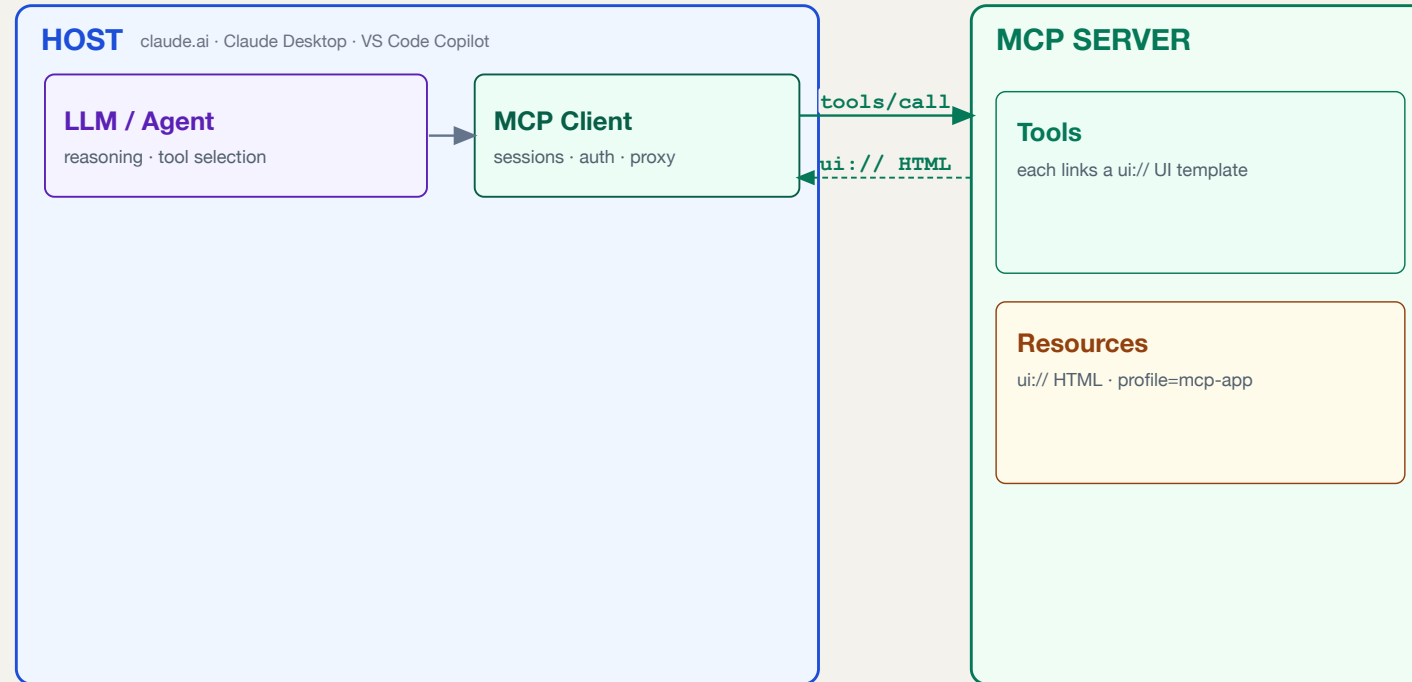
sessions · auth · proxy

SEP-1865 · modelcontextprotocol/ext-apps · 2026-01-26

MCP Server

2 / 4

A tool links a ui:// UI template — calling it returns HTML for the Host to render

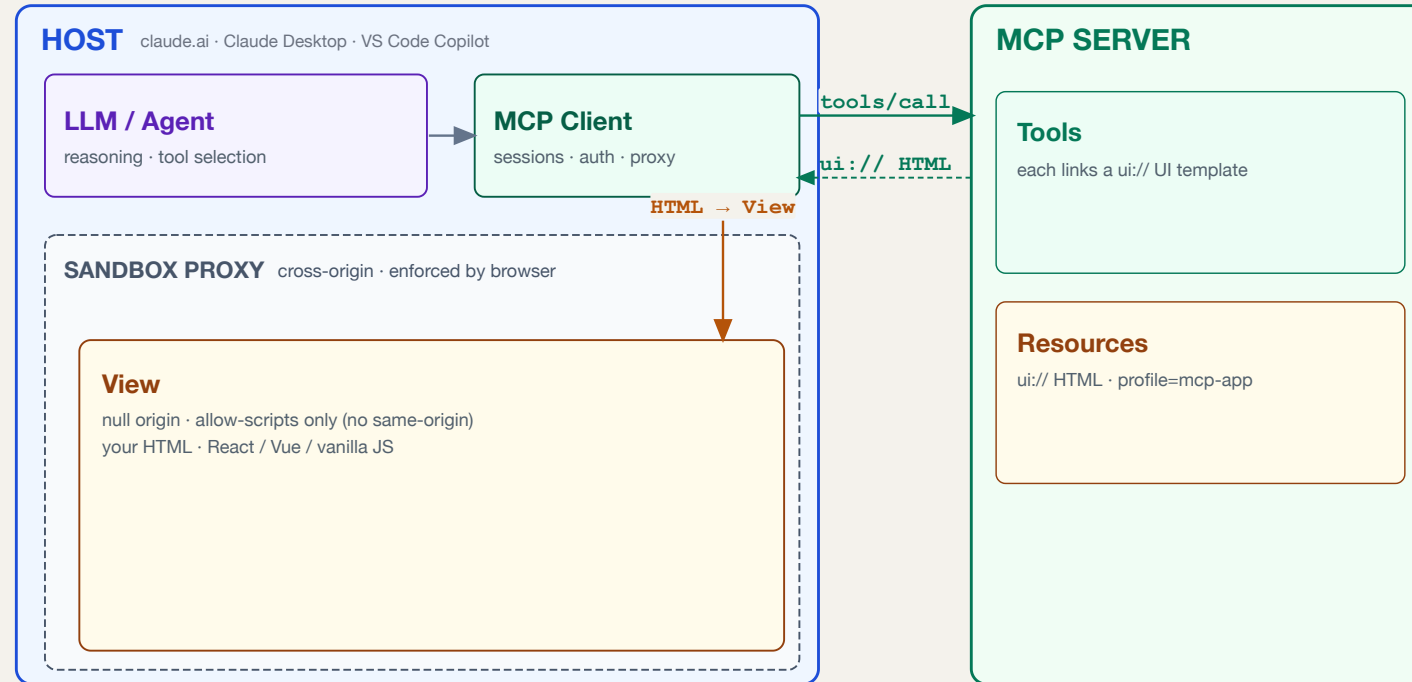


SEP-1865 · modelcontextprotocol/ext-apps · 2026-01-26

Sandbox Isolation

3 / 4

Web hosts wrap the View in a cross-origin sandbox — the browser enforces isolation

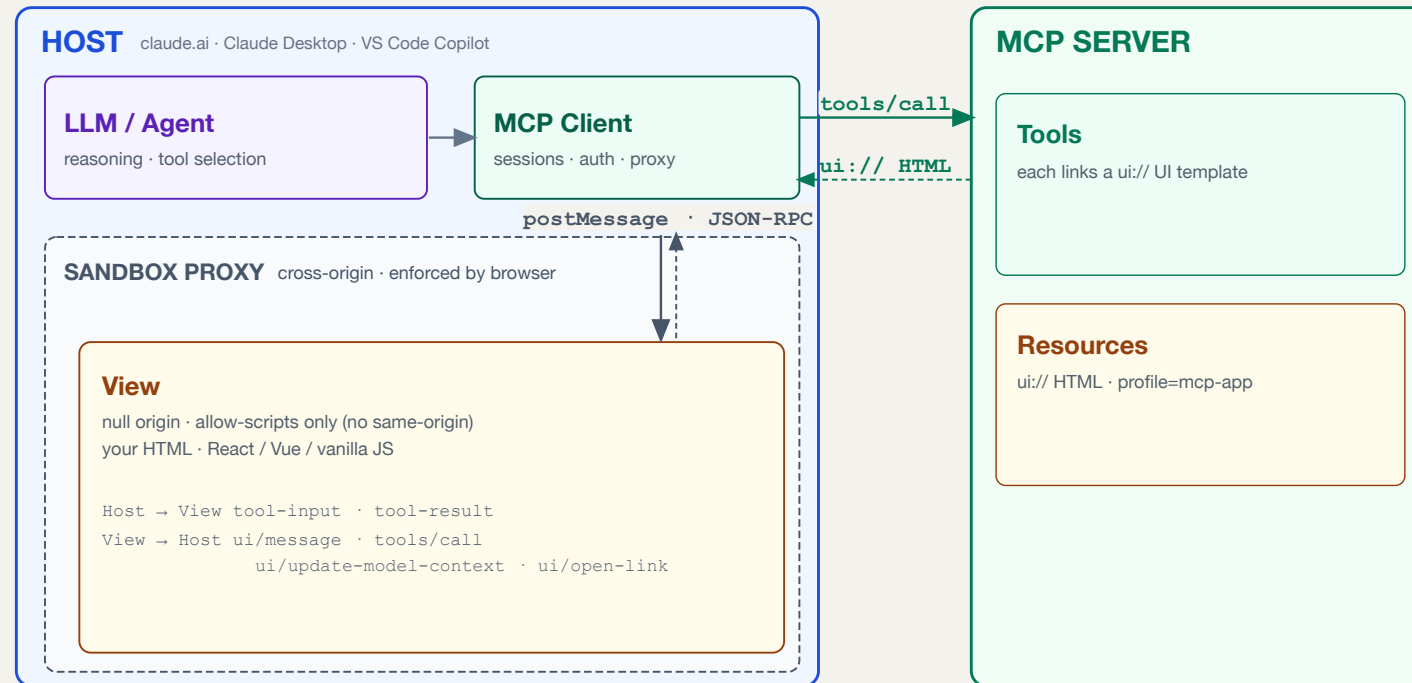


SEP-1865 · modelcontextprotocol/ext-apps · 2026-01-26

Full Protocol Flow

4 / 4

After render, Host and View talk over a bidirectional postMessage bridge (JSON-RPC 2.0)



SEP-1865 · modelcontextprotocol/ext-apps · 2026-01-26

The official dance

Discovery

`tools/list` exposes tool metadata, including `_meta.ui`.

Mount

Host fetches the `ui://` resource and renders the View.

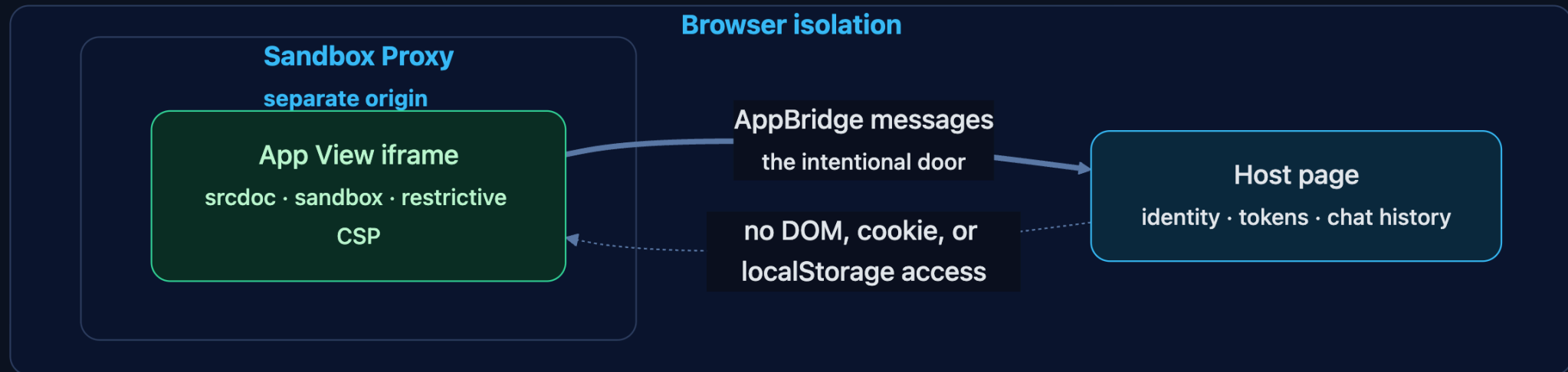
Handshake

View calls `ui/initialize`; Host returns context and capabilities.

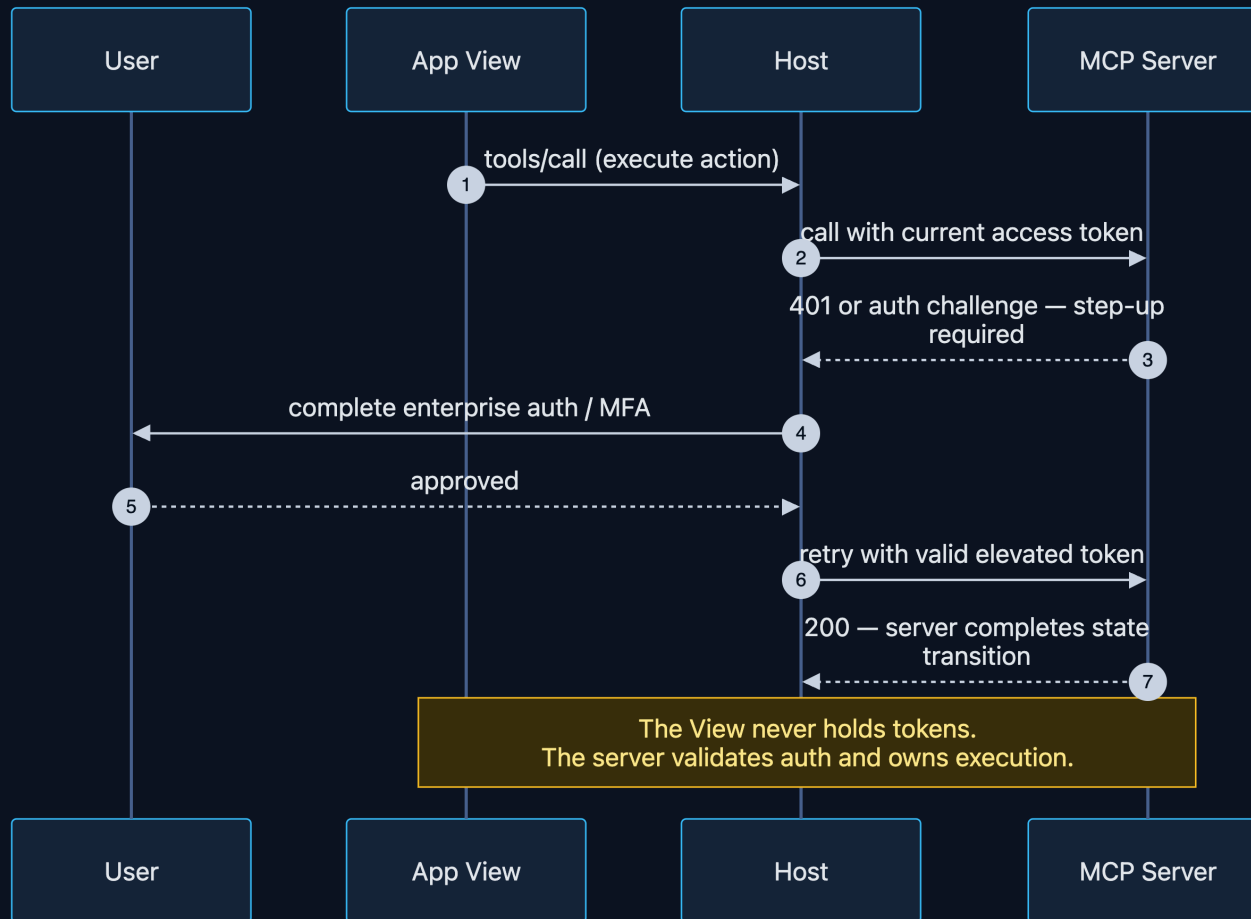
Bridge

JSON-RPC travels over `postMessage` via the AppBridge.

Security · sandboxing & isolation



Authorization · the View never holds tokens



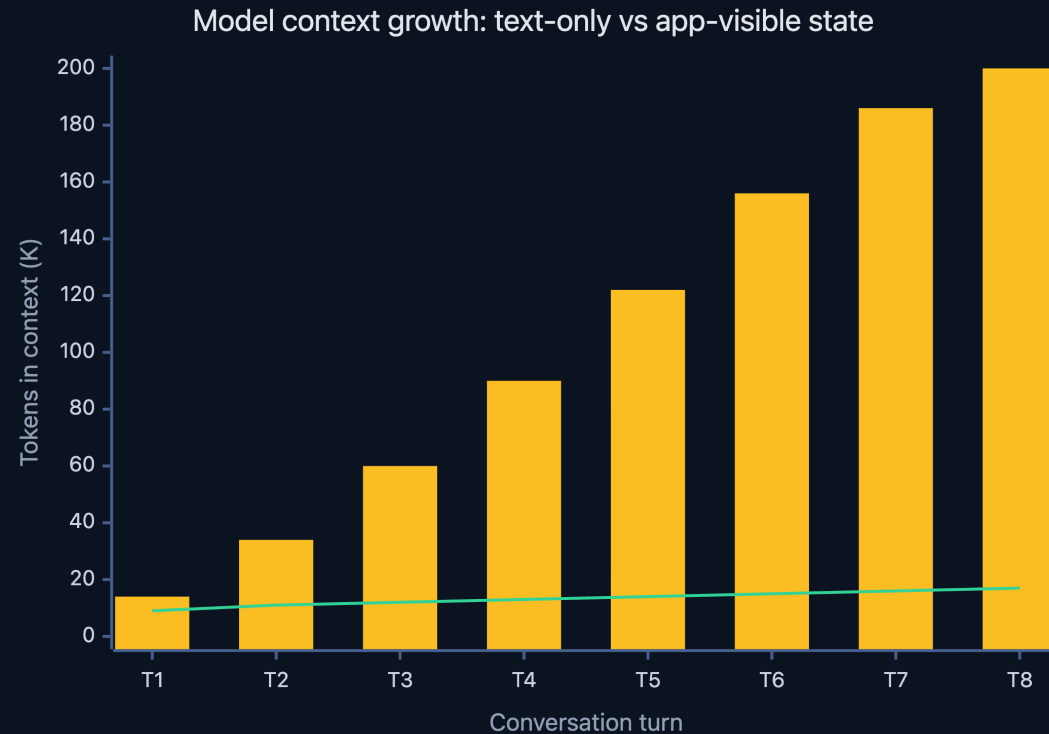
The case to make

Less token spend. More control. Better trust.

1 · less tokens, more personalization

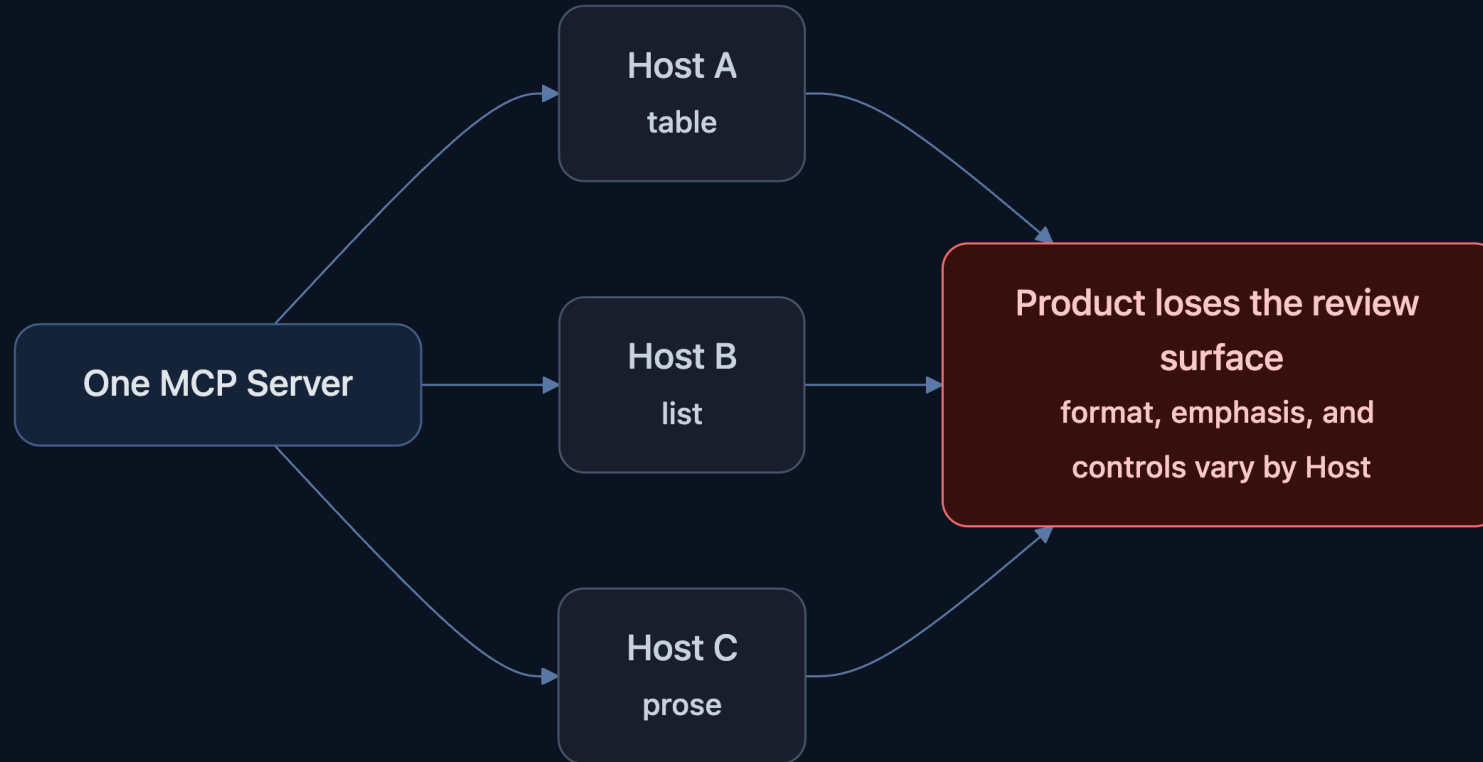


2 • lower cost, fewer hallucinations



Keep heavy data in the View. Keep the model focused on intent and decisions.

3 · trust is a product feature



Supported Hosts

- Claude
- ChatGPT
- VS Code Insiders (desktop)
- Microsoft Copilot
- Goose
- MCP Inspector, MCPJam
- Basic Host (reference host in `modelcontextprotocol/ext-apps`)

Who is Building MCP Apps

- **Dashboards** — Figma, Canva, monday.com
- **HITL confirmations** — Slack workflows

Illustrative examples reported in community discussions — verify per source.

Building MCP Apps

- MCP Apps Implementation:
 - [@modelcontextprotocol/ext-apps](#) SDK by MCP Apps Team
 - [@mcp-ui/server](#) SDK by MCP UI Team
 - Skills
 - [create-mcp-app](#) skill by MCP Apps Team
 - [build-mcp-app](#) skill by Anthropic
 - [mcp-builder](#) by mcp-use Community
- Host Implementation:
 - [@mcp-ui/client](#) SDK by MCP UI Team

References

- [Extension homepage](#)
- [MCP Apps repository](#)
- [SEP-1865 — original proposal](#)
- [MCP Apps proposal post \(Nov 2025\)](#)
- [Official announcement blog post](#)
- [Claude Docs](#)

Security · CSP and CORS

If no CSP metadata is declared, the stable spec requires a restrictive default:

```
default-src 'none';  
script-src 'self' 'unsafe-inline';  
style-src 'self' 'unsafe-inline';  
img-src 'self' data;;  
media-src 'self' data;;  
connect-src 'none';
```

Hosts may further restrict behavior and should log CSP decisions for audit.

declaring CSP

```
{
  text: "<!doctype html>...",
  "_meta": {
    "ui": { "csp": {
      "connectDomains": ["https://www.linuxfoundation.org"], // → connect-src
      "resourceDomains": ["https://aaif.io/"], // → script/style/img/font/media-src
      "frameDomains": ["https://modelcontextprotocol.io"], // → frame-src (else 'none')
      "baseUriDomains": [] // → base-uri (else 'self')
    }
  }
}
```

declaring permissions

```
"_meta": { "ui": {  
  "permissions": {  
    "camera":      {},    // → iframe allow="camera"  
    "microphone":  {},    // → iframe allow="microphone"  
    "geolocation": {},    // → iframe allow="geolocation"  
    "clipboardWrite": {}  // → iframe allow="clipboard-write"  
  }  
}}
```

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Let the model reason.
Let the UI guide.
Let the server enforce.



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